

1 Rollout

s_t a_t r_t $c_{s,t}$

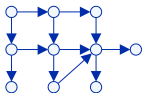


s_{t+1}

Collect transitions
($s_t, a_t, r_t, c_{s,t}, s_{t+1}$)

2 Flow inputs

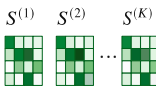
Rollout graph



$$\hat{F} \cdot A \cdot C_k$$



3 Constraints



S



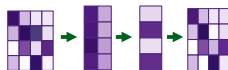
soft: $b_k = B_k/L$
hard: $F_H = 0$

4 Flow projection

$$F^\star = \text{proj}_\epsilon(\hat{F}; b, H)$$

$$\hat{F} \rightarrow F^\star \mid \lambda^\star$$

$K_{ij} = e^{-D_{ij}/\epsilon}$ flow cost target $F^\star$



target flow is not executed

safety prices $\lambda^\star$

5 Priced PPO

$$\lambda = \bar{\lambda}^\star + \beta$$

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priced advantage

$$\tilde{A} = A_r - \lambda \cdot A_c$$

PPO / TRPO

π_{t+1}

